

main

July 24, 2026

```
[1]: num1 = 10_00_000  
     print(num1)
```

1000000

```
[7]: format(0.1, ".25f")
```

```
[7]: '0.10000000000000000055511151'
```

```
[8]: format(0.125, "0.25f")
```

```
[8]: '0.125000000000000000000000000000'
```

```
[9]: 0.1 + 0.1 + 0.1 == 0.3
```

```
[9]: False
```

```
[10]: format(0.1, ".25f")
```

```
[10]: '0.10000000000000000055511151'
```

```
[11]: format(0.3, ".25f")
```

```
[11]: '0.29999999999999999888977698'
```

```
[12]: format(0.2, ".25f")
```

```
[12]: '0.2000000000000000111022302'
```

```
[13]: abs((0.1 + 0.1 + 0.1) - 0.3) < 0.001
```

```
[13]: True
```

```
[14]: 0.125.as_integer_ratio()
```

```
[14]: (1, 8)
```

```
[15]: (10).__add__(25)
```

```
[15]: 35
```

```
[16]: a = float(10)
```

```
[21]: a
```

```
[21]: 10.0
```

```
[26]: c = (-4) ** 0.5
```

```
[27]: c
```

```
[27]: (1.2246467991473532e-16+2j)
```

```
[28]: type(c)
```

```
[28]: complex
```

```
[29]: c.real
```

```
[29]: 1.2246467991473532e-16
```

```
[30]: c.imag
```

```
[30]: 2.0
```

```
[ ]: c = 10 + 2j
```

```
[32]: type(c)
```

```
[32]: complex
```

```
[33]: print(c)
```

```
(10+2j)
```

```
[1]: print("2" < 2)
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[1], line 1  
----> 1 print("2" < 2)  
  
TypeError: '<' not supported between instances of 'str' and 'int'
```

```
[2]: print("2" == 2)
```

```
False
```

```
[12]: a = 0
      b = 1
      c = 500
      d = 0
      e = 1
      f = 500
      print(id(a), id(b), id(c), id(d), id(e), id(f))

11370000 11370032 124619017689520 11370000 11370032 124619017688304
```

```
[13]: 0.1 * 3 - 0.3 < abs(1e-6)
```

[13]: True

```
[16]: a = 257
      a is 257

<>:2: SyntaxWarning: "is" with 'int' literal. Did you mean "=="?
<>:2: SyntaxWarning: "is" with 'int' literal. Did you mean "=="?
/tmp/ipykernel_2767294/274325606.py:2: SyntaxWarning: "is" with 'int' literal.
Did you mean "=="?
      a is 257
```

[16]: False

```
[20]: a = 10
      b = 0
      b != 0 and a / b
```

[20]: False

```
[27]: a = 1e-6
      b = 2e-6
      tol = 1e-7
      print(abs(a - b) <= tol)
```

False

```
[33]: elapsed = 7835
      hours = elapsed // 3600
      minutes = (elapsed - (hours * 3600)) // 60
      seconds = elapsed - ((hours * 3600) + (minutes * 60))
      print(hours, minutes, seconds)
```

2 10 35

```
[34]: elapsed = 7835
      hours = elapsed // (60 * 60)
      minutes = (elapsed % (60 * 60)) // 60
      seconds = (elapsed % (60 * 60)) % 60
```

```
print(hours, minutes, seconds)
```

2 10 35

```
[3]: a = None
      if a == None:
          a = "N/A"
      print(a)
```

N/A

```
[7]: a = None
      a = "N/A" if a is None else a
      print(a)
```

N/A

```
[48]: score = 700
      rating = "Invalid"
      if score < 0:
          rating = "Negative"
      elif score < 580:
          rating = "Poor"
      elif score < 670:
          rating = "Fair"
      elif score < 740:
          rating = "Good"
      elif score < 800:
          rating = "Very Good"
      elif score < 850:
          rating = "Excellent"
      print(rating)
```

Good

```
[52]: elapsed = 1000000
      magnitude = "Invalid"
      if elapsed < 60:
          magnitude = "seconds"
      elif elapsed < (60 * 60):
          magnitude = "minutes"
      elif elapsed < (60 * 60 * 24):
          magnitude = "hours"
      elif elapsed < (60 * 60 * 24 * 7):
          magnitude = "days"
      elif elapsed > (60 * 60 * 24 * 7):
          magnitude = "weeks"
      print(magnitude)
```

weeks